Technical Appendix: Proof of Convergence for Top-k Sparsified AirComp Federated Learning

Jinning Song and Dong Liu

I. Algorithm Flow and Assumptions

To facilitate the analysis, we first formally define the algorithm flow, error feedback mechanism, and core assumptions. Let $r = k/d$ be the sparsification ratio. All vectors are denoted in boldface (e.g., $\boldsymbol{\theta}$, $\mathbf{e}$, $\boldsymbol{\Delta}$).

A. Algorithm Flow and Error Feedback Mechanism

Following the main text, the raw accumulated local update after τ SGD steps is defined as:

$$\boldsymbol{\Delta}_i^t \triangleq \mathbf{u}_i^t = -\eta \sum_{s=1}^{\tau} \mathbf{g}_i^{t,s-1}, \tag{A.1}$$

where $\mathbf{g}_i^{t,s-1}$ is the stochastic gradient satisfying Assumption 2. To compensate for the bias introduced by top-k compression and systemic clipping, each device maintains a local error memory $\mathbf{e}_i^t$, initialized as $\mathbf{e}_i^0 = \mathbf{0}$. The "sparsification-then-clipping" feedback mechanism proceeds as follows:

1) **Compensation:** Form the ideal update $\mathbf{v}_i^t = \boldsymbol{\Delta}_i^t + \mathbf{e}_i^{t-1}$.
2) **Sparsification:** Extract the sparse signal $\tilde{\mathbf{v}}_i^t = \mathrm{Top}_k(\mathbf{v}_i^t)$.
3) **Error Update:** The unselected residual is stored into the memory: $\mathbf{e}_i^t = \mathbf{v}_i^t - \tilde{\mathbf{v}}_i^t$.
4) **Systemic Clipping:** The sparse signal is clipped for transmission: $\mathbf{s}_i^t = \mathrm{clip}(\tilde{\mathbf{v}}_i^t, \eta\tau C)$.

Defining the systemic clipping bias as $\boldsymbol{\xi}_i^t \triangleq \tilde{\mathbf{v}}_i^t - \mathbf{s}_i^t$, we obtain the exact algebraic relationship for the transmitted signal:

$$\mathbf{s}_i^t = \tilde{\mathbf{v}}_i^t - \boldsymbol{\xi}_i^t = \mathbf{v}_i^t - \mathbf{e}_i^t - \boldsymbol{\xi}_i^t = \boldsymbol{\Delta}_i^t + \mathbf{e}_i^{t-1} - \mathbf{e}_i^t - \boldsymbol{\xi}_i^t. \tag{A.2}$$

B. Core Assumptions

Assumption 1 (Smoothness and Lower Bound). *Each local loss function f_i is L-smooth, and the global objective satisfies $f(\boldsymbol{\theta}) \geq f^* > -\infty$ for all $\boldsymbol{\theta} \in \mathbb{R}^d$.*

Assumption 2 (Bounded Gradient Variance). *The variance of the local stochastic gradients satisfies $\mathbb{E}\left\|\nabla f_i(\boldsymbol{\theta}_i^{t,s-1}; \mathcal{B}_i^{t,s}) - \nabla f_i(\boldsymbol{\theta}_i^{t,s-1})\right\|_2^2 \leq \zeta^2$.*

Assumption 3 (Bounded Heterogeneity). *There exist constants $G^2 \geq 1$ and $\kappa^2 \geq 0$ such that $\frac{1}{N}\sum_{i=1}^N \|\nabla f_i(\boldsymbol{\theta})\|_2^2 \leq G^2 \|\nabla f(\boldsymbol{\theta})\|_2^2 + \kappa^2$.*

Assumption 4 (Bounded Clipping Bias). *The system-level clipping bias $\boldsymbol{\xi}_i^t \triangleq \tilde{\mathbf{v}}_i^t - \mathbf{s}_i^t$ satisfies $\mathbb{E}\|\boldsymbol{\xi}_i^t\|_2^2 \leq \eta^2\tau^2\beta^2$ for a constant $\beta > 0$ depending on the clipping threshold C.*

II. Auxiliary Lemmas

Lemma 3 (Contraction Property of Top-k). *For any input vector $\mathbf{v} \in \mathbb{R}^d$, the deterministic compression error satisfies:*

$$\|\mathbf{v} - \mathrm{Top}_k(\mathbf{v})\|_2^2 \leq (1-r)\|\mathbf{v}\|_2^2. \tag{A.3}$$

Proof. Let the coordinates of $\mathbf{v}$ be sorted by magnitude such that $v_{(1)}^2 \geq v_{(2)}^2 \geq \cdots \geq v_{(d)}^2$. The error vector $\mathbf{v} - \mathrm{Top}_k(\mathbf{v})$ consists exactly of the smallest $d-k$ elements. Because these elements have the smallest magnitudes, their average energy is inherently bounded by the average energy of the entire vector:

$$\frac{1}{d-k}\sum_{j=k+1}^{d} v_{(j)}^2 \leq \frac{1}{d}\sum_{j=1}^{d} v_{(j)}^2 = \frac{1}{d}\|\mathbf{v}\|_2^2.$$

Multiplying both sides by $d-k$, we obtain $\|\mathbf{v} - \mathrm{Top}_k(\mathbf{v})\|_2^2 \leq \frac{d-k}{d}\|\mathbf{v}\|_2^2 = (1-r)\|\mathbf{v}\|_2^2$. □

Lemma 4 (Error Contraction Step). *Given $\mathbf{v}_i^t = \boldsymbol{\Delta}_i^t + \mathbf{e}_i^{t-1}$ and $\mathbf{e}_i^t = \mathbf{v}_i^t - \mathrm{Top}_k(\mathbf{v}_i^t)$, the error memory updating satisfies:*

$$\mathbb{E}_t\|\mathbf{e}_i^t\|_2^2 \leq (1-r)\mathbb{E}_t\|\boldsymbol{\Delta}_i^t + \mathbf{e}_i^{t-1}\|_2^2. \tag{A.4}$$

Proof. Taking the expectation on both sides of the deterministic property established in Lemma 3 directly yields $\mathbb{E}_t\|\mathbf{e}_i^t\|_2^2 = \mathbb{E}_t\|\mathbf{v}_i^t - \mathrm{Top}_k(\mathbf{v}_i^t)\|_2^2 \le (1-r)\mathbb{E}_t\|\mathbf{v}_i^t\|_2^2$. Substituting the mechanism definition $\mathbf{v}_i^t = \mathbf{\Delta}_i^t + \mathbf{e}_i^{t-1}$ directly completes the proof. □

Lemma 5 (**Bounded Local Divergence).** *Under Assumptions 1–3, the physical local model drift satisfies:*

$$\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}_t\|\boldsymbol{\theta}^t - \boldsymbol{\theta}_i^{t,s-1}\|_2^2 \le 16\eta^2\tau^2G^2\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 + 16\eta^2\tau^2\kappa^2 + 4\tau\eta^2\zeta^2. \tag{A.5}$$

Proof. According to the local update rule $\boldsymbol{\theta}_i^{t,h} = \boldsymbol{\theta}_i^{t,h-1} - \eta\mathbf{g}_i^{t,h-1}$ (where $h \ge 1$ denotes the local step iteration) and the initial synchronization $\boldsymbol{\theta}_i^{t,0} = \boldsymbol{\theta}^t$, applying the Peter-Paul inequality $\|\mathbf{a}+\mathbf{b}\|_2^2 \le (1+\frac{1}{2\tau-1})\|\mathbf{a}\|_2^2 + 2\tau\|\mathbf{b}\|_2^2$ yields:

$$\begin{aligned}
\mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h} - \boldsymbol{\theta}^t\|_2^2 &= \mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h-1} - \boldsymbol{\theta}^t - \eta\mathbf{g}_i^{t,h-1}\|_2^2 \\
&\le \Big(1+\frac{1}{2\tau-1}\Big)\mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h-1} - \boldsymbol{\theta}^t\|_2^2 + 2\tau\eta^2\mathbb{E}_t\|\mathbf{g}_i^{t,h-1}\|_2^2 \\
&\le \Big(1+\frac{1}{2\tau-1}\Big)\mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h-1} - \boldsymbol{\theta}^t\|_2^2 + 4\tau\eta^2\mathbb{E}_t\|\nabla f_i(\boldsymbol{\theta}_i^{t,h-1}) - \nabla f_i(\boldsymbol{\theta}^t)\|_2^2 + 4\tau\eta^2\mathbb{E}_t\|\nabla f_i(\boldsymbol{\theta}^t)\|_2^2 + \eta^2\zeta_i^2 \\
&\le \Big(1+\frac{1}{2\tau-1} + 4\eta^2L^2\tau\Big)\mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h-1} - \boldsymbol{\theta}^t\|_2^2 + 4\eta^2\tau\mathbb{E}_t\|\nabla f_i(\boldsymbol{\theta}^t)\|_2^2 + \eta^2\zeta_i^2.
\end{aligned}$$

Averaging over all clients $i \in [N]$, applying the dissimilarity Assumption 3 for G^2 and κ^2, and enforcing the learning rate condition $\eta \le \frac{1}{2\sqrt{2}\tau L}$ ensures $4\eta^2L^2\tau \le \frac{1}{2\tau} \le \frac{1}{2\tau-1}$. We obtain:

$$\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h} - \boldsymbol{\theta}^t\|_2^2 \le \Big(1+\frac{1}{\tau-1}\Big)\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}_t\|\boldsymbol{\theta}_i^{t,h-1} - \boldsymbol{\theta}^t\|_2^2 + 4\eta^2\tau G^2\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 + 4\eta^2\tau\kappa^2 + \eta^2\zeta^2.$$

Unrolling the recursion for the local model after $s-1$ updates, we get:

$$\begin{aligned}
\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}_t\|\boldsymbol{\theta}_i^{t,s-1} - \boldsymbol{\theta}^t\|_2^2 &\le \sum_{h=0}^{s-2}\Big(1+\frac{1}{\tau-1}\Big)^h\Big[4\eta^2\tau G^2\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 + 4\eta^2\tau\kappa^2 + \eta^2\zeta^2\Big] \\
&\le (\tau-1)\left[\Big(1+\frac{1}{\tau-1}\Big)^\tau - 1\right] \times \Big[4\eta^2\tau G^2\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 + 4\eta^2\tau\kappa^2 + \eta^2\zeta^2\Big] \\
&\le C_0 \times \Big[4\eta^2\tau^2G^2\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 + 4\eta^2\tau^2\kappa^2 + \tau\eta^2\zeta^2\Big] \\
&\le 16\eta^2\tau^2G^2\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 + 16\eta^2\tau^2\kappa^2 + 4\tau\eta^2\zeta^2,
\end{aligned} \tag{A.6}$$

where C_0 is a universal numerical constant derived from the geometric limit $(1+\frac{1}{\tau-1})^\tau \le e < 3$ (for $\tau > 1$), which is safely bounded by $C_0 \le 4$ to obtain the final clean constants. This holds for any $s \le \tau$, completing the proof. □

Lemma 6 (**Error Memory Bound).** *The global average of the error memory is bounded by:*

$$\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}_t\|\mathbf{e}_i^t\|_2^2 \le \mu\frac{1}{N}\sum_{i=1}^{N}\mathbb{E}_t\|\mathbf{\Delta}_i^t\|_2^2, \tag{A.7}$$

where $\mu = \frac{2(1-r)(2-r)}{r^2}$.

Proof. Using the Peter-Paul inequality $\|\mathbf{a}+\mathbf{b}\|_2^2 \le (1+\alpha)\|\mathbf{a}\|_2^2 + (1+\alpha^{-1})\|\mathbf{b}\|_2^2$ with Lemma 4, we introduce a free parameter $\alpha > 0$ and set $\mathbf{a} = \mathbf{e}_i^{t-1}$ and $\mathbf{b} = \mathbf{\Delta}_i^t$:

$$\mathbb{E}_t\|\mathbf{e}_i^t\|_2^2 \le (1-r)(1+\alpha)\mathbb{E}_t\|\mathbf{e}_i^{t-1}\|_2^2 + (1-r)(1+\alpha^{-1})\mathbb{E}_t\|\mathbf{\Delta}_i^t\|_2^2.$$

To ensure the error term rigorously contracts, we optimally choose $\alpha = \frac{r}{2(1-r)}$. The contraction factor mapping to the past error $\mathbf{e}_i^{t-1}$ becomes $\rho = (1-r)(1+\alpha) = 1 - \frac{r}{2} = \frac{2-r}{2} < 1$. Correspondingly, the coefficient mapping to the update term $\mathbf{\Delta}_i^t$ expands to $\beta_{\mathrm{err}} = (1-r)(1+\alpha^{-1}) = (1-r)\left(1+\frac{2(1-r)}{r}\right) = \frac{(1-r)(2-r)}{r}$. Unrolling the geometric recursion over t steps yields the finite amplification constant $\mu = \frac{\beta_{\mathrm{err}}}{1-\rho} = \frac{(1-r)(2-r)/r}{r/2} = \frac{2(1-r)(2-r)}{r^2}$. □

III. Proof of Main Convergence Bound (Theorem 2)

A. Construction of the Virtual Sequence and Separation

Define the global error memory $\bar{\mathbf{e}}^{t-1} \triangleq \frac{1}{N}\sum_{i=1}^{N}\mathbf{e}_i^{t-1}$ and the virtual global model $\tilde{\boldsymbol{\theta}}^t = \boldsymbol{\theta}^t + \bar{\mathbf{e}}^{t-1}$, which completely absorbs the unsent Top-k errors. From the physical update rule $\boldsymbol{\theta}^{t+1} = \boldsymbol{\theta}^t + \frac{1}{N}\sum_{i=1}^{N}\mathbf{s}_i^t + \frac{\mathbf{z}^t}{Nb_t}$ and the exact structural relation (A.2), the error feedback naturally cancels out, yielding the virtual recursion:

$$\tilde{\boldsymbol{\theta}}^{t+1} = \tilde{\boldsymbol{\theta}}^t + \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{\Delta}_i^t - \bar{\boldsymbol{\xi}}^t + \frac{\mathbf{z}^t}{Nb_t}, \tag{A.8}$$

where $\bar{\boldsymbol{\xi}}^t = \frac{1}{N}\sum_{i=1}^{N}\boldsymbol{\xi}_i^t$. Applying the L-smoothness descent lemma yields the separation:

$$\mathbb{E}_t[f(\tilde{\boldsymbol{\theta}}^{t+1}) - f(\tilde{\boldsymbol{\theta}}^t)] \le \underbrace{\mathbb{E}_t\langle\nabla f(\tilde{\boldsymbol{\theta}}^t), \frac{1}{N}\sum_i \boldsymbol{\Delta}_i^t - \bar{\boldsymbol{\xi}}^t\rangle}_{T_1} + \underbrace{\frac{L}{2}\mathbb{E}_t\Big\|\frac{1}{N}\sum_i \boldsymbol{\Delta}_i^t - \bar{\boldsymbol{\xi}}^t + \frac{\mathbf{z}^t}{Nb_t}\Big\|_2^2}_{T_2}. \tag{A.9}$$

B. Bounding T_1 and T_2

1) Descent Direction T_1: Substituting $\boldsymbol{\Delta}_i^t = -\eta\sum_{s=1}^{\tau}\mathbf{g}_i^{t,s-1}$ and applying the unbiasedness of gradients, the primary gradient inner product formulation becomes:

$$\mathbb{E}_t\langle\nabla f(\tilde{\boldsymbol{\theta}}^t), \frac{1}{N}\sum_i \boldsymbol{\Delta}_i^t\rangle = -\eta\tau\mathbb{E}_t\langle\nabla f(\tilde{\boldsymbol{\theta}}^t), \frac{1}{N\tau}\sum_{i,s}\nabla f_i(\boldsymbol{\theta}_i^{t,s-1})\rangle.$$

To decouple the global gradient from the local heterogeneous gradients, we apply the inner product identity $\langle a, b\rangle = \frac{1}{2}\|a\|_2^2 + \frac{1}{2}\|b\|_2^2 - \frac{1}{2}\|a-b\|_2^2$. This yields:

$$-\frac{\eta\tau}{2}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 - \frac{\eta\tau}{2}\mathbb{E}_t\Big\|\frac{1}{N\tau}\sum_{i,s}\nabla f_i(\boldsymbol{\theta}_i^{t,s-1})\Big\|_2^2 + \frac{\eta\tau}{2}\mathbb{E}_t\Big\|\frac{1}{N\tau}\sum_{i,s}\big(\nabla f(\tilde{\boldsymbol{\theta}}^t) - \nabla f_i(\boldsymbol{\theta}_i^{t,s-1})\big)\Big\|_2^2.$$

Applying Jensen's inequality and L-smoothness to the last term produces the heterogeneity drift $\frac{\eta L^2}{2N}\sum_{i,s}\mathbb{E}_t\|\tilde{\boldsymbol{\theta}}^t - \boldsymbol{\theta}_i^{t,s-1}\|_2^2$. For the separate clipping bias term, applying the Peter-Paul inequality $-\langle a, b\rangle \le \frac{\eta\tau}{4}\|a\|_2^2 + \frac{1}{\eta\tau}\|b\|_2^2$ alongside Assumption 4 guarantees:

$$-\mathbb{E}_t\langle\nabla f(\tilde{\boldsymbol{\theta}}^t), \bar{\boldsymbol{\xi}}^t\rangle \le \frac{\eta\tau}{4}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + \frac{1}{\eta\tau}(\eta^2\tau^2\beta^2) = \frac{\eta\tau}{4}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + \eta\tau\beta^2.$$

Summing the expanded terms, the primary gradient coefficient becomes $-\frac{\eta\tau}{2} + \frac{\eta\tau}{4} = -\frac{\eta\tau}{4}$, which we safely relax to $-\frac{\eta\tau}{8}$ to ensure ample descent margin. Hence:

$$T_1 \le -\frac{\eta\tau}{8}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + \frac{\eta L^2}{2N}\sum_{i,s}\mathbb{E}_t\|\tilde{\boldsymbol{\theta}}^t - \boldsymbol{\theta}_i^{t,s-1}\|_2^2 - \frac{\eta\tau}{2}\mathbb{E}_t\Big\|\frac{1}{N\tau}\sum_{i,s}\nabla f_i(\boldsymbol{\theta}_i^{t,s-1})\Big\|_2^2 + \eta\tau\beta^2. \tag{A.10}$$

2) Variance and Noise T_2: To handle the variance components, we note that the privacy-preserving channel noise relies solely on its real part, i.e., $\mathrm{Re}\{n_m^t\} \sim \mathcal{N}(0, \sigma_0^2/2)$, implying $\mathbb{E}_{z^t}\|\mathbf{z}^t\|_2^2 = d\frac{\sigma_0^2}{2}$. Thus, separating the zero-mean channel noise ($\mathbb{E}_{z^t}[\mathbf{z}^t] = \mathbf{0}$) and sequentially applying the basic inequality $\|\mathbf{a}-\mathbf{b}\|_2^2 \le 2\|\mathbf{a}\|_2^2 + 2\|\mathbf{b}\|_2^2$, Jensen's inequality, and Assumption 4, we obtain:

$$T_2 \le L\frac{1}{N}\sum_i\mathbb{E}_t\|\boldsymbol{\Delta}_i^t\|_2^2 + L\eta^2\tau^2\beta^2 + \frac{L\sigma_0^2 d}{4N^2 b_t^2}. \tag{A.11}$$

Expanding $\mathbb{E}_t\|\boldsymbol{\Delta}_i^t\|_2^2$ completely via Assumptions 2 and 3 yields the explicit form:

$$T_2 \le 2L\eta^2\tau^2G^2\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + \frac{2L^3\eta^2\tau}{N}\sum_{i,s}\mathbb{E}_t\|\tilde{\boldsymbol{\theta}}^t - \boldsymbol{\theta}_i^{t,s-1}\|_2^2 + L\eta^2\tau^2(2\kappa^2 + \zeta^2) + L\eta^2\tau^2\beta^2 + \frac{L\sigma_0^2 d}{4N^2 b_t^2}. \tag{A.12}$$

C. Bounding the Virtual Local Divergence

Define the local divergence toward the virtual model as $\Psi_t \triangleq \frac{1}{N}\sum_{i,s}\mathbb{E}_t\|\tilde{\boldsymbol{\theta}}^t - \boldsymbol{\theta}_i^{t,s-1}\|_2^2$. Using the algebraic decomposition $\tilde{\boldsymbol{\theta}}^t - \boldsymbol{\theta}_i^{t,s-1} = \bar{\mathbf{e}}^{t-1} + (\boldsymbol{\theta}^t - \boldsymbol{\theta}_i^{t,s-1})$ and Lemma 5, together with the gradient conversion bound $\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 \le 2\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + 2L^2\|\bar{\mathbf{e}}^{t-1}\|_2^2$, we obtain:

$$\Psi_t \le 64\eta^2\tau^3G^2\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + 2\tau(1 + 32\eta^2\tau^2G^2L^2)\mathbb{E}_t\|\bar{\mathbf{e}}^{t-1}\|_2^2 + 32\eta^2\tau^3\kappa^2 + 8\tau^2\eta^2\zeta^2. \tag{A.13}$$

D. Learning Rate Conditions and Single-Step Descent

Combining T_1 and T_2, substituting Ψ_t, and safely bounding the global error memory $\mathbb{E}_t\|\bar{\mathbf{e}}^{t-1}\|_2^2$ via Lemma 6 (using Jensen's inequality, $\|\bar{\mathbf{e}}^{t-1}\|_2^2 \le \frac{1}{N}\sum_{i=1}^N \|\mathbf{e}_i^{t-1}\|_2^2$), the total positive gradient penalty accumulates to:

$$C_\nabla = 2L\eta^2\tau^2G^2 + \eta L^2\Big(\frac{1}{2} + 2L\eta\tau\Big)\Big[64\eta^2\tau^3G^2 + 4\mu\eta^2\tau^3G^2(1+32\eta^2\tau^2G^2L^2)\Big]. \tag{A.14}$$

For guaranteed convergence, we strictly enforce $C_\nabla \le \frac{\eta\tau}{16}$, which decomposes into the following sufficient conditions:

1) $2L\eta\tau G^2 \le \frac{1}{32} \;\Rightarrow\; \eta \le \frac{1}{64L\tau G^2}$,
2) $\frac{1}{2} + 2L\eta\tau \le 1 \;\Rightarrow\; \eta \le \frac{1}{4L\tau}$,
3) $32\eta^2\tau^2G^2L^2 \le 0.5 \;\Rightarrow\; \eta \le \frac{1}{8\tau GL}$,
4) $L^2\eta^2\tau^2G^2(64+6\mu) \le \frac{1}{32} \;\Rightarrow\; \eta \le \frac{1}{\tau GL\sqrt{2048+192\mu}}$.

Noting that $G \ge 1$ holds by Assumption 3, it is straightforward to verify that $\frac{1}{64L\tau G^2} < \min\left\{\frac{1}{4L\tau}, \frac{1}{8\tau GL}\right\}$. Therefore, conditions 2 and 3 are always strictly subsumed by condition 1. Consequently, the effective learning rate is governed purely by the bottleneck of conditions 1 and 4, which directly yields the simplified two-term constraint presented in the main text:

$$\eta \le \min\left\{\frac{1}{64L\tau G^2}, \frac{1}{\tau GL\sqrt{2048+192\mu}}\right\}.$$

Under these exact constraints, the one-round descent formula simplifies elegantly to:

$$\mathbb{E}_t[f(\tilde{\boldsymbol{\theta}}^{t+1}) - f(\tilde{\boldsymbol{\theta}}^t)] \le -\frac{\eta\tau}{8}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + \eta^2\tau^2LC_{\mathrm{var}}(\kappa^2+\zeta^2) + \mathcal{E}_{\mathrm{clip}} + \frac{L\sigma_0^2 d}{4N^2b_t^2}, \tag{A.15}$$

where $\mathcal{E}_{\mathrm{clip}} \triangleq (\eta\tau + L\eta^2\tau^2)\beta^2$, and the intrinsic heterogeneity constant C_{var} is defined precisely as (assuming $\kappa^2+\zeta^2 > 0$; otherwise variance identically vanishes):

$$C_{\mathrm{var}} \triangleq \frac{2\kappa^2+\zeta^2}{\kappa^2+\zeta^2} + \eta L\Big[\frac{32\tau\kappa^2+8\zeta^2}{\kappa^2+\zeta^2} + 2\tau\mu(1+32\eta^2\tau^2G^2L^2)\frac{2\kappa^2+\zeta^2}{\kappa^2+\zeta^2}\Big]. \tag{A.16}$$

E. Virtual Model Convergence

Telescoping the sum over $t = 0, \ldots, T-1$ and recognizing $f(\boldsymbol{\theta}^0) = f(\tilde{\boldsymbol{\theta}}^0)$, we define the initial distance $\Delta f \triangleq f(\boldsymbol{\theta}^0) - f^*$. Rearranging the equation yields the tight virtual bound:

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 \le \frac{8\Delta f}{T\eta\tau} + 8\eta\tau LC_{\mathrm{var}}(\kappa^2+\zeta^2) + \frac{8\mathcal{E}_{\mathrm{clip}}}{\eta\tau} + \frac{2L\sigma_0^2 d}{T\eta\tau N^2}\sum_{t=0}^{T-1}\frac{1}{b_t^2}. \tag{A.17}$$

F. Conversion to the Physical Model

From the conversion $\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 \le 2\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + 2L^2\mathbb{E}_t\|\bar{\mathbf{e}}^{t-1}\|_2^2$, we explicitly define the error sequence $E_{t-1} \triangleq 2L^2\mathbb{E}_t\|\bar{\mathbf{e}}^{t-1}\|_2^2$. Invoking Lemma 6 (with Jensen's inequality) and substituting the exact bound of Ψ_{t-1}, we establish a rigorous recursive contraction relation for the error sequence:

$$E_{t-1} \le 8L^2\mu\eta^2\tau^2G^2\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^{t-1})\|_2^2 + 3L^2\mu\eta^2\tau^2(2\kappa^2+\zeta^2) + \rho E_{t-2}, \tag{A.18}$$

where the contraction factor is defined as $\rho \triangleq 6L^2\mu\eta^2\tau^2$. The rigorous learning rate constraints derived above strictly guarantee $\rho \le \frac{1}{32} \le 0.5$. Unrolling this recursion over T rounds and applying the bounding logic for geometric series sum $\frac{1}{1-\rho} \le 2$, we eliminate the recursive term:

$$\begin{aligned}\sum_{t=0}^{T-1}E_{t-1} &\le \frac{1}{1-\rho}\sum_{t=0}^{T-1}\Big(8L^2\mu\eta^2\tau^2G^2\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + 3L^2\mu\eta^2\tau^2(2\kappa^2+\zeta^2)\Big)\\ &\le 16L^2\mu\eta^2\tau^2G^2\sum_{t=0}^{T-1}\mathbb{E}_t\|\nabla f(\tilde{\boldsymbol{\theta}}^t)\|_2^2 + 6L^2\mu\eta^2\tau^2T(2\kappa^2+\zeta^2).\end{aligned} \tag{A.19}$$

Substituting this sequence sum back gives the physical gradient amplification factor $\Gamma \triangleq 2 + 16L^2\mu\eta^2\tau^2G^2$. Scaling (A.17) entirely by Γ and correctly assigning the residual error-delay constants produces the final physical convergence bound:

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}_t\|\nabla f(\boldsymbol{\theta}^t)\|_2^2 \le \frac{2\Gamma L\sigma_0^2 d}{T\eta\tau N^2}\sum_{t=0}^{T-1}\frac{1}{b_t^2} + \Phi(k), \tag{A.20}$$

$$\Phi(k) \triangleq \frac{8\Gamma\Delta f}{T\eta\tau} + \frac{8\Gamma\mathcal{E}_{\mathrm{clip}}}{\eta\tau} + 8\Gamma\eta\tau LC_{\mathrm{var}}(\kappa^2+\zeta^2) + 6L^2\mu\eta^2\tau^2(2\kappa^2+\zeta^2), \tag{A.21}$$

which completes the proof of Theorem 2.